EXP NO: 3
DATE: 30.07.26

TO STUDY THE PACKET TRACER TOOL INSTALLATION AND USER INTERFACE OVERVIEW

Aim:

To understand environment of CISCO PACKET TRACER to design simple network.

Introduction:

A simulator, as the name suggests, simulates network devices and its environments, packet tracer is an exciting network design, simulation and modelling tool.

1. It allows you to model complex systems without the need for dedicated equipment
2. It helps you to model complex systems practice your network configuration and troubleshooting skills via computer or an android or ios based mobile device
3. It is available for both the linux and windows desktop environment
4. Protocols in Packet Tracer are coded to work and behave in the same way as they would on real hardware.

INSTALLING PACKET TRACER:

To download Packet Tracer, go to https://www.netacad.com and login with your Cisco Networking Academy credentials; then, click on the Packet Tracer graphic and download the package appropriate for your operating system.

WINDOWS

Installation in windows is pretty simple and straight forward; the setup comes in a single .file named packet tracer - setup 6.0.1.exe open the file to begin the setup wizard, accept the licence agreement, choose a location, and start the installation

LINUX:

Linux users with an ubuntu/debian distribution should download the file for ubuntu, and those using Fedora/Redhat/centos must download the file for Fedora. Grant executable permission to this file by using chmod, and execute it to begin the installation.
chanod +x, packet tracer 601-i386-installer-rpm.bin.

./Packet Tracer 601-i386-installer-rpm.bin

USER INTERFACE OVERVIEW:

The layout of the packet Tracer is divided into several components. The components of the Packet Tracer interface are as follows, match the numbering with explanations.

1. menu bar - This is common menu found in all software applications; it is used to open, save, print, change preferences and so on.

2. Main toolbar - This bar provides shortcut icons to menu options that are commonly accessed such as open, save, zoom, undo, and redo and on the right hand side is an icon for entering network information for the current networks.

3. Logical/physical workspace tabs - These tab allow you to toggle between the logical and physical work areas.

4. Workspace - This is the area where the topology are created and simulations are displayed.

5. common tools bar - this toolbar provides control for manipulating topologies, such as selects, move layout, place note, delete, inspect, resize shape and add simple/complex P.D.U.

6. Real-time/simulation tabs - these tabs are used to toggle between the real and simulation mode. Buttons are also provided to control the time and to capture the packet.

7. Network component box - This component contains all of the network and end devices available with packet tracer, and is further divided into two areas: Area 7a : Device type selection box - this area contains device categories Area 7b : Device-specific selection box displays the different device models within that category.

8. User-controlled packet box - users can create highly customized packets to test their topology from this area, and the results are displayed as a list

d) Analyse the behaviour of network devices using CISCO PACKET TRACER simulator.

1. From the network component box, click and drag and drop the below components.
   a) 4 Generic PCs and one HUB
   b) 4 Generic PCs and one switch

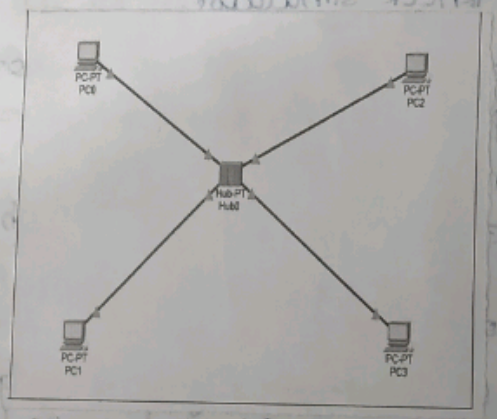
2. Click on connections:
   a) Click on Copper-straight-through cable,
   b) Select one of the PC and connect it to HUB using the cable. The Link LED should glow in green, indicating that the link is up. Similarly connect remaining 3 PCs to the HUB.
   c) Similarly connect 4 PCs to the switch using copper straight through cable.

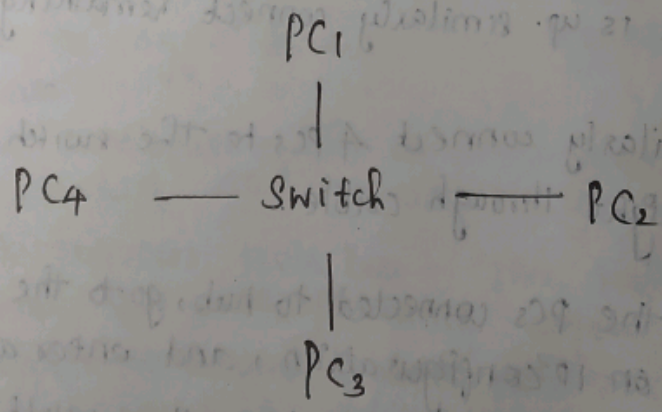
3. Click on the PCs connected to hub, go to the Desktop tab, click on IP configuration, and enter an IP address and subnet mask. Here the default gateway and DNS server information is not needed as there are only two end devices in the network.

4. Observe the flow of PDU from source PC to destination PC by selecting the Real Time mode of simulation.

5. Repeat step #3 to step #5 for the PCs connected to the switch

6. Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviors of switch and HUB.





Result:

Hence the environment of CISCO PACKET TRACER to design simple network was understood